

SIGNAL & DATA ANALYSIS IN NEUROSCIENCE
2018
PSTH AND JPSTH

Ayala Matzner

biu.sigproc@gmail.com

Outline

- PSTH
- JPSTH

PSTH – peri-stimulus time histogram

- “Peri” = Around
- Purpose: provide information on activity (e.g. spikes) with respect to a specific event in time (stimulus)

PSTH

Scheme for aligning spikes to an event
(e.g. sensory spikes relative to a sensory cue):

- Align spike sequences with the multiple stimuli onset.
- Divide the observation period T into bins of size Δt .
- Count the number of spikes in each bin for every stimuli.
- Draw a bar-graph histogram with the average spike count normalized to rate (divide by Δt).

$$PSTH(t = k \cdot \Delta t) = \frac{E[\text{nSpikes in bin } k]}{\Delta t} \left[\frac{\text{spikes}}{\text{sec}} \right]$$

JPSTH- joint PSTH

• Definition: A co-incidence matrix - Gives information regarding co-activation of 2 neurons with relation to a specific event.

• Construction:

- Align the 2 spike sequences with the onset of stimuli.
Repeat for n trials (trials).

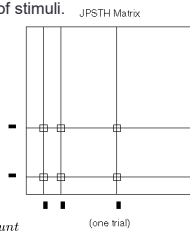
- Divide the observation period T into N bins of size Δt . (same as PSTH)

• For each trial:

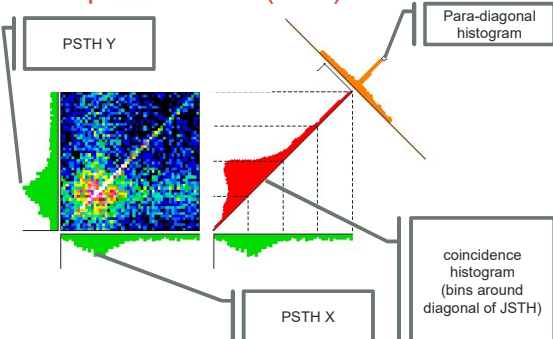
if t_x is time of a spike of X and
 t_y is time of a spike of Y
Increment $JPSTH(t_x, t_y)$ by 1

- Repeat for all trials (on the same JPSTH)

- The count PSTH can be normalized to probability by:
$$PSTH_{prob} = \frac{PSTH_{count}}{n \text{ trials}}$$



Example - JPSTH (cont)



Shift predictor

- Purpose: It is important to remove correlations that arise due to co-stimulation:
Since we stimulate both cells at the same time, spike trains of the cells may co-vary, and we are not interested in co-variance that is due to co-stimulation
- To overcome this – we construct a **shift predictor**, a matrix representing all this co-variance, and then subtract it from the JPSTH

Shift predictor

- **Shift Predictor Construction 1** (long way):
 - Same as raw JPSTH, but repeat with trials shifted (e.g. align trial 3 of neuron x with trial 4 of neuron y).
 - Repeat n times (all possible shifts..)
 - Normalize to count: divide by: 'n Shifts preformed'.
 - Normalize to probability: divide by: '(n Shifts preformed * #Trials)'
- **Shift Predictor Construction 2** (short way):
 - Cross product of PSTHs (that are already shuffled by definition):
 $\text{shiftPredictor} = \text{PSTHy} * \text{PSTHx}$; where
 $\text{size}(\text{PSTHy}) = \text{size}(\text{PSTHx}) = [\text{Nbins}, 1]$
- **Corrected JPSTH** = JPSTH – shiftPredictor
 Note: perform with JPSTH, Shift Predictor normalized to probability.

Example - Corrected PSTH

